

EXPERIMENT 2

DEVELOP AND COMPARE CLI, GUI, AND VOICE USER INTERFACES (VUI)

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AIM:

To design and develop applications using GUI, CUI, and VUI interfaces for understanding different user interaction methods.

PROCEDURE:

i) CLI (Command Line Interface)

```
tasks = []
```

```
def add_task(task):
```

```
    tasks.append(task)
```

```
    print(f"Task '{task}' added.")
```

```
def view_tasks():
```

```
    if tasks:
```

```
        print("\nYour Tasks:")
```

```
        for idx, task in enumerate(tasks, 1):
```

```
            print(f"{idx}. {task}")
```

```
    else:
```

```
        print("No tasks to show.")
```

```
def remove_task(task_number):
```

```
    if 0 < task_number <= len(tasks):
```

```
        removed_task = tasks.pop(task_number - 1)
```

```
        print(f"Task '{removed_task}' removed.")
```

```

else:
    print("Invalid task number.")
def main():
    while True:
        print("\nOptions:")
        print("1. Add Task")
        print("2. View Tasks")
        print("3. Remove Task")
        print("4. Exit")
        choice = input("Enter your choice: ")
        if choice == '1':
            task = input("Enter task: ")
            add_task(task)
        elif choice == '2':
            view_tasks()
        elif choice == '3':
            task_number = int(input("Enter task number to remove: "))
            remove_task(task_number)
        elif choice == '4':
            print("Exiting...")
            break
        else:
            print("Invalid choice. Please try again.")
if __name__ == "__main__":
    main()

```

OUTPUT:

Options:

1. Add Task
2. View Tasks
3. Remove Task
4. Exit

Enter your choice: 1

Enter task: Complete assignment

Task 'Complete assignment' added.

Enter your choice: 2

Your Tasks:

1. Complete assignment

ii) GUI (Graphical User Interface)

```
import tkinter as tk
```

```
from tkinter import messagebox
```

```
tasks = []
```

```
def add_task():
```

```
    task = task_entry.get()
```

```
    if task:
```

```
        tasks.append(task)
```

```
        task_entry.delete(0, tk.END)
```

```
        update_task_list()
```

```
    else:
```

```
        messagebox.showwarning("Warning", "Task cannot be empty")
```

```
def update_task_list():
```

```

    task_list.delete(0, tk.END)

    for task in tasks:

        task_list.insert(tk.END, task)

def remove_task():

    selected_task_index = task_list.curselection()

    if selected_task_index:

        task_list.delete(selected_task_index)

        tasks.pop(selected_task_index[0])

app = tk.Tk()

app.title("To-Do List")

task_entry = tk.Entry(app, width=40)

task_entry.pack(pady=10)

add_button = tk.Button(app, text="Add Task", command=add_task)

add_button.pack(pady=5)

remove_button = tk.Button(app, text="Remove Task", command=remove_task)

remove_button.pack(pady=5)

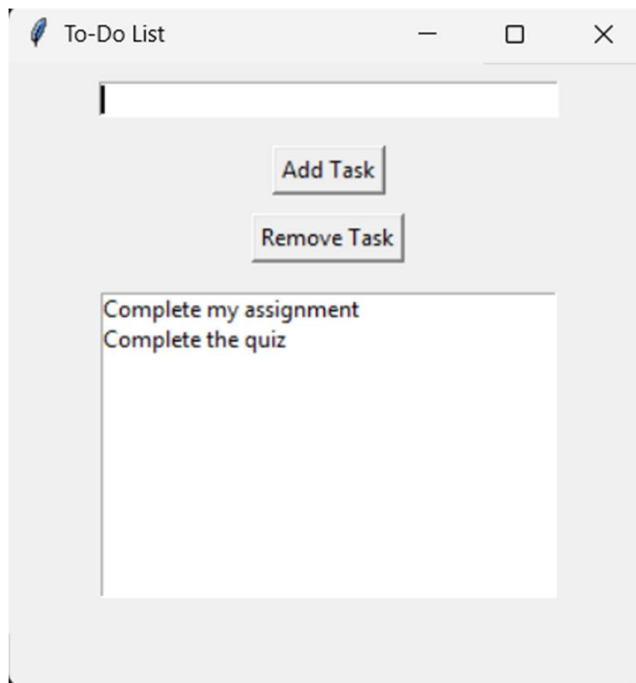
task_list = tk.Listbox(app, width=40, height=10)

task_list.pack(pady=10)

app.mainloop()

```

OUTPUT:



iii) VUI (Voice User Interface)

```
import speech_recognition as sr
import pyttsx3
tasks = []
recognizer = sr.Recognizer()
engine = pyttsx3.init()
def add_task(task):
    tasks.append(task)
    engine.say(f"Task {task} added")
    engine.runAndWait()
def view_tasks():
    if tasks:
        engine.say("Your tasks are")
        for task in tasks:
            engine.say(task)
```

```

else:
    engine.say("No tasks to show")
    engine.runAndWait()
def remove_task(task_number):
    if 0 < task_number <= len(tasks):
        removed_task = tasks.pop(task_number - 1)
        engine.say(f"Task {removed_task} removed")
    else:
        engine.say("Invalid task number")
    engine.runAndWait()
def recognize_speech():
    with sr.Microphone() as source:
        print("Listening...")
        audio = recognizer.listen(source)
    try:
        command = recognizer.recognize_google(audio)
        return command
    except sr.UnknownValueError:
        engine.say("Sorry, I did not understand that")
        engine.runAndWait()
        return None
def main():
    while True:
        engine.say("Options: add task, view tasks, remove task, or exit")
        engine.runAndWait()
        command = recognize_speech()

```

```

if not command:
    continue
if "add task" in command:
    engine.say("What is the task?")
    engine.runAndWait()
    task = recognize_speech()
    if task:
        add_task(task)
elif "view tasks" in command:
    view_tasks()
elif "remove task" in command:
    engine.say("Which task number to remove?")
    engine.runAndWait()
    task_number = recognize_speech()
    if task_number:
        remove_task(int(task_number))
elif "exit" in command:
    engine.say("Exiting...")
    engine.runAndWait()
    break
else:
    engine.say("Invalid option. Please try again.")
    engine.runAndWait()
if __name__ == "__main__":
    main()

```

OUTPUT:

System Voice:

"Options: add task, view tasks, remove task, or exit"

User Says:

"Add task"

System Voice:

"What is the task?"

User Says:

"Finish project report"

System Voice:

"Task Finish project report added"

RESULT

The GUI, VUI and CLI To-Do List applications successfully managed tasks using the respective python commands.